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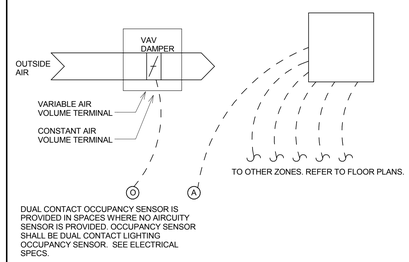
DEMAND CONTROL VENTILATION WITH CENTRALIZED CO2 SENSOR

PROJECT DESCRIPTION	POINT NAME	BI	BO	BV	AI	AO	AV	ALARM TYPE	GRAPHIC
OCCUPIED SCHEDULE	OCC_SCHED						X		YES
UNOCCUPIED SCHEDULE	UNOCC_SCHED						X		YES
VAV SET POINT MINIMUM OCCUPIED	VAV_SP_MIN_OCC					X			YES
VAV SET POINT MAXIMUM OCCUPIED	VAV_SP_MAX_OCC					X			YES
VAV SET POINT STANDBY OCCUPIED	VAV_SP_STBY_OCC					X			YES
VAV SET POINT UNOCCUPIED	VAV_SP_UNOCC					X			YES
VAV OCC SENSOR OCCUPIED	VAV_OCCS_OCC	X							
VAV OCC SENSOR UNOCCUPIED	VAV_OCCS_UNOCC	X							
VAV CFM ACTUAL	VAV_CFM_ACTUAL				X				YES
VAV DAMPER %OPEN	VAV_DAM_%				X				YES
ZONE CO2 LEVEL (PPM)	CO2_PPM				X				YES
ZONE CO2 SET POINT	CO2_SP					X			YES
CHANGE CO2 SENSOR ALARM	CO2_S_ALARM	X						BOOLEAN COMMAND FAIL	YES

ALL SPACES SERVED BY VARIABLE AIR VOLUME TERMINAL OR CONSTANT AIR VOLUME TERMINAL



DUAL CONTACT OCCUPANCY SENSOR IS PROVIDED IN SPACES WHERE NO AIRCUITY SENSOR IS PROVIDED. OCCUPANCY SENSOR SHALL BE DUAL CONTACT LIGHTING OCCUPANCY SENSOR. SEE ELECTRICAL SPECS.

DEMAND CONTROL VENTILATION WITH CENTRALIZEDCO2 SENSOR(AIRCUITY)

1. DEMAND CONTROL VENTILATION SEQUENCE OF OPERATION (BASIS OF DESIGN: AIRCUITY)

1.0 OCCUPIED MODE:

1.1. Aircuity FMS shall monitor CO2 levels in each zone.

1.2. Aicuity FMS shall compare the CO2 levels in each zone with OA CO2. A differential CO2 signal shall be sent to the BMS. The signal shall be a 0-10 VDC signal where 0 VDC represents 200 PPM differential, 5 VDC represents 500 PPM and 10 VDC represents 800 PPM differential.

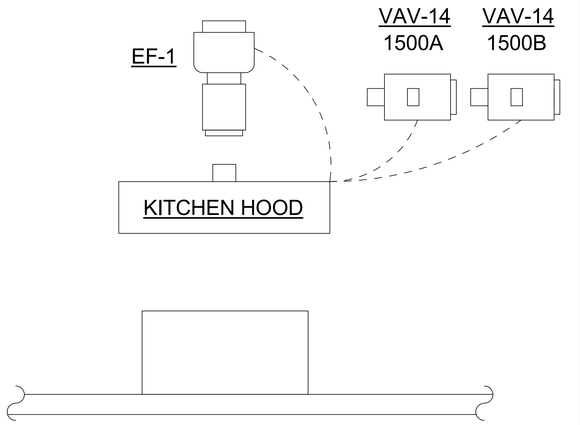
1.3. The VAV box controller shall modulate the VAV box damper to maintain as follows: 200 PPM (adj.) damper is closed, 500 PPM damper is half open (adj.), and 800 PPM (adj.) damper is full open.

1.4. Controls contractor is responsible for providing field mounted VAV box controller to accomplish sequence and achieve all listed control points.

1.5 If a pace is not provided with an aircuity sensor, VAV box shall respond to occuapancy sensor. Durring occpied hours the VAV box shall have an occpied standby CFM setting that will enable when NO occupancy is detected and an occupied CFM setting that will enable when occupancy is detected. If VAV box serves more than one space then all space occupany sensors must detect NO occupancy before VAV box is allowed to enable occupied standyby CFM setting.

MECHANICAL CONTROL LEGEND

AFF	ABOVE FINISHED FLOOR	Ts	
BAS	BUILDING AUTOMATION SYSTEM		
CO_2	CARBON DIOXIDE	H	
TCC	TEMPERATURE CONTROL CONTRACTOR	LL	
DP	DEWPOINT		
EA	EXHAUST AIR PATH	P	
RA	RETURN AIR PATH		
SA	SUPPLY AIR PATH	DP	
HPS/R	HEAT PUMP WATER SUPPLY/RETURN	DPSW	
NC	NORMALLY CLOSED		
OA	OUTSIDE AIR PATH	DPS	
OCC	OCCUPANCY	C	
PRESS	PRESSURE		
		M	
DI	DIGITAL INPUT		
DO	DIGITAL OUTPUT	F	
AI	ANALOG INPUT		
AO	ANALOG OUTPUT	CS	
VFD	VARIABLE FREQUENCY DRIVE	SD	AVERAGING TEMPERATURE SENSOR
RH	RELATIVE HUMIDITY		
MAU	MAKE-UP AIR UNIT	COS	AVERAGING TEMPERATURE SENSOR
O/H	OCCUPIED HEATING SETPOINT	DSP-HL	AVERAGING TEMPERATURE SENSOR
O/C	OCCUPIED COOLING SETPOINT		
U/H	UNOCCUPIED HEATING SETPOINT	DSP-LL	AVERAGING TEMPERATURE SENSOR
U/C	UNOCCUPIED COOLING SETPOINT		
K	EMERGENCY HVAC/VENTILATION KILL BUTTON	ZN-DP	AVERAGING TEMPERATURE SENSOR
Ts	AVERAGING TEMPERATURE SENSOR	ZN-CO2	AVERAGING TEMPERATURE SENSOR
		ZN-OCC	AVERAGING TEMPERATURE SENSOR
		ZN-T	AVERAGING TEMPERATURE SENSOR
		CO2	AVERAGING TEMPERATURE SENSOR
		O	CENTRIFUGAL FAN



KITCHEN HOOD & EXHAUST CONTROL SEQUENCE

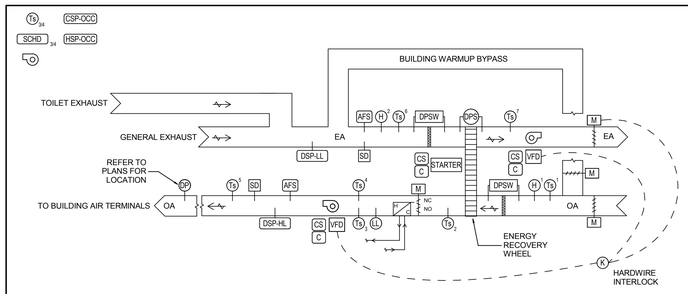
NOT TO SCALE

KITCHEN HOOD & EXHAUST CONTROL SEQUENCES:

1. The kitchen exhuast hood shall be interlocked with Exhuast Fan EF-1, Gym Air Terminal Box VAV-14 1500A and VAV-14 1500B, and normally open Gymnasium 1500 motorized exhaust damper serving louver L-2.
2. Upon activation of EF-1, VAV-14 1500A and 1500B shall both be activated and each set to a minimum of 1500 CFM. Both boxes shall return to normal operation upon the deactivation of exhaust fan EF-1.
3. Upon activation of EF-1 Normally open Gymnasium 1500 motorized exhaust damper shall be energized closed so that gym over pressurizes and transfers air for make-up into the kitchen. Exhaust damper shall return to normal position upon deactivation of EF-1.

KITCHEN HOOD & EXHAUST CONTROL POINTS

POINT DESCRIPTION	POINT NAME	BI	BO	BV	AI	AO	AV	ALARM TYPE	GRAPHICS
KITCHEN HOOD STATUS	DW_S	X							X
EXHAUST FAN COMMAND	DW_EF_C		X						X
VAV BOX COMMAND	DW_VAV_C		X						X
MOTORIZED EXHAUST DAMPER	EA_D_C		X						X
MOTORIZED EAXHAUST DAMPER STATUS	EA_D_S	X							X



OUTSIDE AIR SYSTEM (OA-1) SCHEMATIC

NOT TO SCALE

OUTSIDE AIR SYSTEMS:

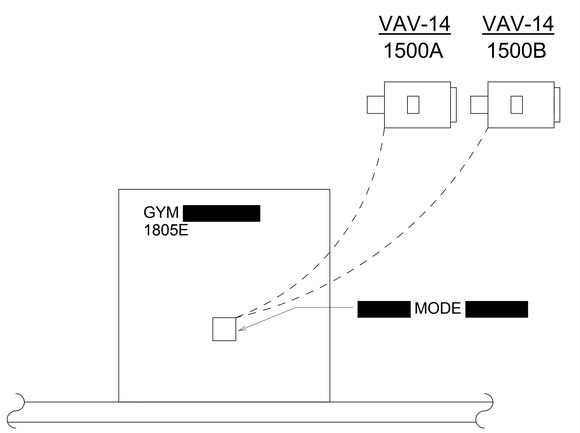
1. THERE ARE TWO OUTSIDE AIR UNITS IN THIS FACILITY.
2. EACH SYSTEM SHALL OPERATE UNDER THE CONTROL OF A LOCAL, STAND-ALONE MICROPROCESSOR BASED DDC CONTROLLER.
3. EACH SYSTEM SHALL BE PLACED INTO THE OCCUPIED/UNOCCUPIED MODE BASED UPON THE USER ADJUSTABLE SCHEDULE AT THE NETWORK CONTROLLER. THESE SYSTEMS SHALL BE IN THE OCCUPIED MODE DURING REGULAR SCHOOL HOURS ONLY. SEE BUILDING OCCUPANCY SCHEDULE.
4. IF COMMUNICATION IS LOST BETWEEN THE NETWORK CONTROLLER AND THE OUTISDE AIR SYSTEM CONTROLLER, THEN THE OUSIDE AIR SYSTEM SHALL BE PLACED INTO THE UNOCCUPIED MODE UNTIL COMMUNICATION IS RESTORED.
5. EACH SYSTEM WILL BE PLACED INTO A MODE OF OPERATION BASED UPON THE FOLLOWING ADJUSTABLE TERMPERATURE SCHEDULE:

OUTSIDE AIR TEMPERATURE	MODE OF OPERATION
75 DEG F OR GREATER (ADJ.)	COOLING MODE
BETWEEN 50 DEG F AND 75 DEG F (ADJ.)	ECONOMIZER MODE
50 DEG F OR LESS (ADJ.)	HEATING MODE

6. IN THE OCCUPIED MODE OF FREEZESTAT (LOW LIMIT) MODE:
   - THE SUPPLY FAN AND EXHAUST SHAL BE OFF.
   - THE ENERGY RECOVERY WHEEL SHALL BE OFF.
   - THE OUTSIDE AIR DAMPER AND EXHAUST AIR DAMPER SHALL BE FULLY CLOSED,
   - FACE AND BYPASS DAMPER SHALL BE IN FULL FACE POSITION.
7. WHEN PLACED INTO THE OCCUPIED MODE, THE FOLLOWING SHALL OCCUR IN SEQUENTIAL ORDER AFTER OPERATION OF THE WATER-TO-WATER HEAT PUMP SYSTEM HAS BEEN PROVED:
   - THE ENERGY RECOVERY WHEEL SHALL START AND OPERATION SHALL BE PROVED VIA CURRENT SWITCH,
   - THE OUTSIDE AIR DAMPER AND EXHAUST AIR DAMPER SHALL FULLY OPEN AND BE PROVED VIA CURRENT SWITCH,
   - THE SUPPLY FAN/VFD AND EXHAUST FAN/VFD SHALL START AND OPERATION SHALL BE PROVED VIA CURRENT SWITCH,
   - THE SYSTEM SHALL NOT START IF ANY ONE COMPONENT DOES NOT PROVE OPERATION, INCLUDING THE HYDRONIC WATER-TO-WATER HEAT EXCHANGER SYSTEM.
8. IN THE OCCUPIED MODE, THE FACE AND BYPASS DAMPERS SHALL MODULATE TO MAINTAIN DISCHARGE AIR TEMPERATURE (ADJ.) BASED UPON THE FOLLOWING SCHEDULE:

DISCHARGE AIR TEMPERATURE	MODE OF OPERATION
68 DEG F (ADJ.)	COOLING MODE
VARIES	ECONOMIZER MODE
65 DEG F (ADJ.)	HEATING MODE
(NOTE THAT THERE SHOULD BE NOT BE MORE THAN APPROXIMATELY 60% AIRFLOW NECESSARY ACROSS THE COOLING/HEATING COIL TO MAINTAIN REQUIRED DISCHARGE AIR TEMPERATURE. SET DAMPER ACTUATOR CONTROL ACCORDINGLY.)	

9. IF THE OUTSIDE AIR TEMPERATURE IS BETWEEN 60 DEG F (ADJ.) AND 65 DEG F (ADJ.), THEN THE ENERGY RECOVERY WHEEL SHALL BE OFF. THE WATER - TO -WATER HEAT PUMP SYSTEM SHALL REMAIN OFF.
10. DEHUMIDIFICATION MODE: WHEN THE DUCT-MOUNTED EXHAUST AIR RELATIVE HUMIDITY SENSOR READS 65% OR GREATER, ENERGY RECOVERY WHEEL SHALL REMAIN ON AND BYPASS DAMPER SHALL CLOSE ALLOWING THE UNIT TO OPERATE WITH HALF AIRFLOW UNTIL EXHAUST AIR RELATIVE HUMIDITY IS BELOW 60%.
11. THE SUPPLY FAN SHALL BE CONTROLLED BY THE VFD BASED ON FIELD-MOUNTED DUCT PRESSURE SETPOINT. FAN SUPPLY STATIC PRESSUREOPTIMIZATION SHALL BE UTILIZED BY POLLING OF ASSOCIATED VAV/CAV BOX DAMPER POSITIONS AND RESETTING OF PRESSURE SETPOINT AsALLOWED. DDC SYSTEM DETERMINES VAV BOX WITH GREATEST DAMPER OPEN POSITION ONCE EVERY TEN MINUTES. UNIT'S SUPPLY AIR STAtPRESSURE SETPOINT DECREASED BY 0.1" WG AND 1.25" WG. SETPOINT IS 0.75" WG WHEN UNIT IS FIRST STARTED. DETERMINE IN THE FIELD WITH THE TAB CONTRCTOR THE STATIC PRESSURE SETPOINT TO OBTAIN ACCEPTABLE AIRFLOW AS DESIGNED.
12. HARDWARE INTERLOCK SUPPLY FAN, EXHAUST FAN, AND ISOLATION DAMPERS WITH EMERGENCY HVAC/VENTILATION KILL BUTTON.
13. THE EXHAUST FAN SHALL TRACK THE OUTSIDE AIR FAN V.A AIRFLOW STATION AND SHALL MAINTAIN A 1,000 CFM OFFSET FOR EACH UNIT TO MAINTAIN A POSITIVE BUILDING PRESSURIZATION. ALL RESTROOMS ARE PROVIDED WITH A CONSTANT VOLUME BOX AND SHALL MAINTAIN THE CFM INDICATED ON THE DRAWINGS. IF WHIL THE EXHAUST FAN TRACKS THE OUTSIDE AIR FAN THE CFM IN THE RESTROOM DROPS BELOW THIS [redacted] IN ANY ROOM BY 10%, THEN THE [redacted] AIR [redacted] STOP [redacted] TO [redacted] AIR FLOW.
14. THE COIL IS A FULL FLOW COIL WITH FACE AND [redacted] FOR [redacted] AND [redacted] CONTROL. LOW [redacted] IS [redacted] ON THE [redacted] OF THE COIL AND [redacted] WHEN THE [redacted] AIR [redacted] 35 DEG F. THE [redacted] AND [redacted] FANS [redacted] STOP, [redacted] AIR AND [redacted] AIR [redacted] CLOSE.
15. LOW [redacted] ERW [redacted] ON THE [redacted] OF THE [redacted] WHEEL, WHEN THE [redacted] AIR [redacted] 40 DEG F THE [redacted] AND [redacted] FANS [redacted] STOP, [redacted] AND [redacted] AIR [redacted] CLOSE. WHEN [redacted] TO 45 DEG F (ADJ.) [redacted] TO [redacted] OPERATION.
16. A [redacted] LOW [redacted] OF THE CHILLED/HOT WATER COIL [redacted] STOP THE [redacted] OF THE [redacted] IF THE [redacted] 35 DEG F.
17. A [redacted] BE [redacted] IN EACH AIR SYSTEM. IF [redacted] IS DETECTED, THEN THE [redacted] AND AN AUDIO/VISUAL [redacted] ACTIVATE. UPON [redacted] OF PROBLEM, THE [redacted] BE [redacted] AND [redacted] TO [redacted] OPERATION. [redacted] WITH FIRE [redacted] SYSTEM. ANY [redacted] OF BUILDING'S FIRE [redacted] THE OA [redacted] COMPLETELY.
18. THE UNIT [redacted] BE [redacted] WITH A [redacted] WARM UP/COOL DOWN [redacted] DUCT. THIS [redacted] THE [redacted] AND [redacted] A [redacted] AND NOT USE ANY [redacted] AIR [redacted] THE [redacted] IS UNOCCUPIED. IF THE [redacted] HAS [redacted] THE FANS [redacted] BE [redacted] AND THE UNIT. [redacted] 95 DEG F (ADJ.). THE [redacted] VAV BOX IN EACH ZONE [redacted] OPEN [redacted] THE [redacted] AIR UNIT TO [redacted] TEMPERATURE. CONVERSELY, IF THE [redacted] SETPOINT, THE UNIT [redacted] TO A 55 DEG F (ADJ.) [redacted] AIR [redacted] TO [redacted] THE [redacted] TEMPERATURE. THE [redacted] VAV BOX IN EACH ZONE [redacted] OPEN. ALL [redacted] BE [redacted] WARMUP/COOL DOWN CYCLE.



NOT TO SCALE

[redacted] AIR [redacted] SEQUENCES:

1. [redacted] 1800 **normal mode** of operation shall be such that outside air is provided for classroom use. [redacted] 1800 normal minimum shall be 1200 CFM and normal maximum shall 1600 CFM. VAV-box A serving gym shall maintain CFM values and modulate according to CO2.
2. [redacted] 1800 **event mode** shall be initiated by a push button in Gym Storage 1805E and start a 2 hour timer (adj.) for an event mode which will automically disable upon experation of timer. [redacted] 1800 Event mode shall have a minimum CFM value of 1600 CFM and maximum value of 3000 CFM. VAV box A and B shall work together and modulate based on CO2 to maintain CFM values.
3. Note that Kitchen hood operation shall overide both of the above modes and take controls of VAV-boxes A and B serving gym.

OA-1 POINTS LIST

DISPLAY NAME	POINT NAME	BI	BO	BV	AI	AO	AV	ALARM TYPE	GRAPHIC
BUILDING STATIC PRESSURE	BLDG_SP				X			OUT OF RANGE	YES
DUAL TEMP MODULATING 2-WAY %OPEN	DT_CV-1_PO					X		BOOLEAN COMMAND FAIL	YES
DUAL TEMP MODULATING 2-WAY OVERIDE	DT_CV-1_OVRD						X		
DEHUMIDIFICATION SETPOINT	OA_DHUM_SPT						X		YES
OA INTAKE HUMIDITY SENSOR 1	OA_HUM_S_1				X				
EA DISCHARGE HUMIDITY SENSOR 2	EA_HUM_S_2				X				
OA_EA BYPASS DAMPER	OA_EA_BYP_DAMP_C					X		BOOLEAN COMMAND FAIL	
OA_EA BYPASS DAMPER STATUS	OA_EA_BYP_DAMP_ST	X							
OA_EA BYPASS DAMPER OVERIDE	OA_EA_BYP_DAMP_OVRD						X		YES
OA [redacted]	OA_INT_DAMP_C					X		[redacted] FAIL	YES
OA [redacted]	OA_INT_DAMP_ST	X							YES
OA [redacted]	OA_INT_DAMP_OVRD						X		YES
EA [redacted]	EA_INT_DAMP_C					X		[redacted] FAIL	YES
EA [redacted]	EA_INT_DAMP_ST	X							YES
EA [redacted]	EA_INT_DAMP_OVRD						X		YES
[redacted] AIR [redacted] 1	OA_TEMP_S_1				X				YES
[redacted] AIR [redacted] 2	OA_TEMP_S_2				X				YES
[redacted] AIR H/C [redacted] 3	OA_TEMP_S_3				X				YES
[redacted] AIR [redacted] 4	OA_TEMP_S_4				X				YES
[redacted] AIR [redacted] 5	OA_TEMP_S_5				X				
[redacted] AIR T3/T4 [redacted]	OA_TEMP_S_AVG_3&4						X		
OA [redacted]	OA_FLTR_DP_S				X			OUT OF [redacted]	
OA [redacted] 1	OA_RH_S				X				YES
[redacted] FACE AND [redacted]	OA_FBYP_DAMPER					X			YES
LOW [redacted] STAT	LL-FRZ_STAT	X						OUT OF [redacted]	YES
SCHEDULE	OCC_S			X					
OA FANS START/STOP	OAFAN_RUN_C		X					[redacted] FAIL	YES
OA FANS START/STOP [redacted]	OAFAN_RUN_OVRD			X				[redacted] FAIL	YES
OA FANS [redacted]	OAFAN_RUN_ST	X						[redacted] FAIL	YES
OA FANS VFD [redacted]	OAFAN_VFDFLT_S	X						[redacted] OF [redacted]	YES
OA FANS VFD [redacted]	OAFAN_VFD_SPD_C					X			YES
OA FANS VFD [redacted]	OAFAN_VFD_SPD_OVRD						X		YES
FAN 1 OA AIR CFM	OAFAN1_AF_CFM				X				YES
FAN 2 OA AIR CFM	OAFAN2_AF_CFM				X				YES
OA [redacted] CFM	OA_AF_CFM_TOTAL				X				YES
OA [redacted]	OA_SMOKE_DET	X						[redacted] OF [redacted]	YES
OA [redacted]	OA_SP_S				X				YES
OA [redacted]	OA_SP_SETP						X		YES
OA [redacted]	OA_SP_RESET_SETP						X		YES
[redacted]	OCC_CLG_SETP						X		YES
[redacted]	OCC_HTG_SETP						X		YES
DUCT [redacted] HIGH [redacted]	OA_DSP_HL	X						OUT OF [redacted]	YES
DUCT [redacted] LOW [redacted]	EA_DSP_LL	X						OUT OF [redacted]	
[redacted]	UNOCC_WARMUP_HTG_SETP						X		
EA [redacted]	EA_SMOKE_DET	X						[redacted] FAIL	
[redacted] AIR [redacted] 6	EA_TEMP_S_6				X				
[redacted] AIR [redacted] 7	EA_TEMP_S_6				X				YES
EA [redacted]	EA_FLTR_DP_S				X			OUT OF [redacted]	YES
[redacted] DIFF. PRESS. [redacted]	ERW_DP_S				X			OUT OF [redacted]	
[redacted] START/STOP [redacted]	ERW_RUN_C		X					[redacted] FAIL	
[redacted] START/STOP [redacted]	ERW_RUN_OVRD			X				[redacted] FAIL	
[redacted]	ERW_RUN_ST	X						[redacted] FAIL	
[redacted]	ERW_FRZ_C			X					
EA FANS START/STOP	EAFAN_RUN_C							[redacted] FAIL	
EA FANS START/STOP [redacted]	EAFAN_RUN_OVRD							[redacted] FAIL	
EA FANS [redacted]	EAFAN_RUN_ST							[redacted] FAIL	
EA FANS VFD [redacted]	EAFAN_VFDFLT_S							[redacted] OF [redacted]	
EA FANS VFD [redacted]	EAFAN_VFD_SPD_C								
EA FANS VFD [redacted]	EAFAN_VFD_SPD_OVRD								
FAN 1 EA AIR CFM	EAFAN1_AF_CFM								
FAN 2 EA AIR CFM	EAFAN2_AF_CFM								
EA [redacted] CFM	EA_AF_CFM_TOTAL								
SUM OF [redacted]	TERM_UNIT_CFM_TOTAL				X				

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